
Nikhil Cheriya

Phoenix, AZ 85022 | +16029350455 | ntej18.ui@gmail.com
Portfolio: [Nikhil Cheriya](#)

Summary

Senior Full Stack Software Engineer with 8+ years of experience building **production-grade** backend services and **AI platform integrations** using **TypeScript, Node.js, Python, and React**. Recently architected and shipped a **Model Context Protocol (MCP)** tool integrating with **Anthropic Claude** and **OpenAI ChatGPT**, enabling seamless inline restaurant booking within AI chat interfaces **delivered MVP to production in 3 months**. Deep expertise in **designing and evolving production APIs, LLM function-calling pipelines, and scalable system architecture** with a proven track record across fintech and payments platforms generating \$1.5M+ in revenue. Experienced operating in **AWS and GCP cloud environments**, contributing design and delivery decisions within small, **high-ownership teams**, and leveraging **AI-assisted development tools** to drive productivity and innovation.

Skills

- **Backend & APIs:** Node.js, Python, Express, TypeScript, REST APIs, GraphQL, WebSockets, MCP (Model Context Protocol), LLM function-calling pipelines, production-grade API design, service interface design, Authentication (OAuth2 / OIDC / Okta)
 - **Frontend:** React, Next.js, JavaScript (ES6+), Context API, Reducer pattern, Redux, Hooks, Tailwind CSS, HTML5, CSS3, Design Systems (Amex DLS)
 - **Databases & Data:** PostgreSQL, MySQL, MSSQL, Couchbase, relational data modeling, schema design, production schema migrations, BigQuery, Redis, Google Cloud Storage (GCS)
 - **Cloud & DevOps:** AWS (EC2, S3, Lambda, RDS), GCP, Docker, GitHub Actions, Jenkins, CI/CD Pipelines, scalable system design
 - **AI & Emerging Tech:** MCP tool development, LLM integration (Anthropic Claude, OpenAI ChatGPT), AI platform integrations, function-calling pipelines, GitHub Copilot, AI-assisted development tools
 - **Observability & Tooling:** ELK Stack (Elasticsearch, Logstash, Kibana), Jest, React Testing Library, Cypress, Playwright, Webpack, Vite, ESLint, SonarQube
 - **Analytics, Performance & Accessibility:** Adobe Analytics, AEM, SEO, Core Web Vitals, WCAG 2.1/2.2, Axe DevTools, Lighthouse
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Career Highlights

- Architected and shipped a **Model Context Protocol (MCP)** tool integrating with **Anthropic Claude** and **OpenAI ChatGPT** platforms, enabling inline restaurant booking via a **React/TypeScript** widget directly within AI chat interfaces **delivered MVP to production in 3 months** with a **Python**-based MCP server, **Couchbase** data store, and **ELK** observability stack.

- Designed and evolved **production-grade REST and GraphQL service interfaces** using **Node.js** and **TypeScript**, applying **backward-compatible schema evolution** and **API versioning strategies** to support high-traffic payment and recommendation workflows.
 - Architected scalable backend services with deliberate **architectural tradeoffs** selecting Redis for low-latency caching over DB reads and GCS for cost-efficient document storage improving **system scalability** and reducing API latency by 40%.
 - Designed and implemented a personalized recommendation engine delivering dine-in and hotel recommendations using transaction history and spending behavior, driving a 50% increase in recommendation relevance and a 32% improvement in click-through rates.
 - Managed **relational database data models** and **production schema design** across **PostgreSQL, MySQL, MSSQL**, and **Couchbase**, ensuring data integrity and query performance at scale.
 - Strengthened quality, reliability, and security by implementing Cypress/Playwright E2E testing, **CI/CD automation** (GitHub Actions, Jenkins), **ELK-based distributed tracing**, and SonarQube vulnerability scanning, ensuring **production-ready** releases.
 - Leveraged **AI-assisted development tools** including **GitHub Copilot** and hands-on **LLM platform integrations** (Anthropic, OpenAI) to accelerate development velocity, code review, and iterative problem-solving.
 - Partnered with product, design, and business stakeholders to lead design and delivery decisions within small, **high-ownership teams**, delivering features in Agile sprints with 99% on-time delivery.
 - Mentored junior engineers on **TypeScript/Node.js** best practices, **production API design**, and testing strategies, reducing onboarding time by 40%.
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Experience

SENIOR SOFTWARE ENGINEER
American Express - Phoenix, AZ

Feb 2026 - Present

- Architected and deployed a **Model Context Protocol (MCP)** tool enabling **AI platforms** (Anthropic Claude, OpenAI ChatGPT) to surface a **React/TypeScript** restaurant booking widget inline within the chat interface, delivering a seamless end-to-end reservation experience without leaving the conversation window — **shipping the initial MVP to production in just 3 months** from inception to launch.
- Engineered a **Python-based MCP server** exposing structured tool definitions consumed by **LLM function-calling pipelines**, allowing models to invoke real-time **Resy** restaurant availability and reservation APIs based on natural language user prompts.
- Designed and implemented a **Couchbase** document store to persist reservation session state, user preferences, and restaurant availability data, enabling fast lookups and resilient booking workflows across high-concurrency AI chat sessions.
- Integrated **ELK (Elasticsearch, Logstash, Kibana)** observability stack for **end-to-end distributed tracing and structured logging** across MCP tool invocations, enabling real-time monitoring, latency profiling, and rapid incident diagnosis across AI platform integrations.

SENIOR SOFTWARE ENGINEER
American Express - Phoenix, AZ

Jan 2025 - Feb 2026

- Designed and evolved **production-grade REST and GraphQL service interfaces** using **Node.js, TypeScript**, and **Express**, with versioning and **backward-compatible API design** supporting 200+ internal stakeholders and high-volume card member workflows.
- Architected backend services with intentional **scalability tradeoffs** — leveraging Redis for sub-millisecond caching over **relational database** reads and **GCS** for document storage — reducing API response latency by 40%.

- Built **production data models** in **PostgreSQL / GCS**, designing schemas to support recommendation engine outputs, model thresholds, and business rule configurations.
- Operated and monitored services on **AWS** and **GCP** cloud platforms, ensuring **production system reliability** and scalability under peak load.
- Leveraged **AI-assisted tools** including **GitHub Copilot** to accelerate development velocity, code review, and iterative problem-solving across the full-stack.
- Led full-stack development using React, Context API, Reducer pattern, custom hooks, and Amex DLS components to build responsive, accessible forms and data tables.
- Achieved 100% WCAG 2.1 AA compliance, reducing accessibility defects in production to near zero.

SENIOR SOFTWARE DEVELOPER
American Express - Phoenix, AZ

May 2023 - Dec 2024

- Led **production-grade API** development and frontend architecture for Card-on-File and partner merchant platforms (Amazon, PayPal, Macy's), supporting high-volume **production** payment workflows contributing to \$1.5M+ annual revenue.
- Designed **scalable system architecture** for partner integrations, evaluating **architectural tradeoffs** between monolithic and service-oriented patterns to support extensibility across 6+ merchant platforms.
- Managed **relational database data models** and query optimization across **PostgreSQL** and **MySQL** to support real-time KPI tracking and reporting workflows.
- Modernized frontend architecture using React, **TypeScript**, Next.js, and Vite, reducing build times by 25%, improving bundle size by 18%, and increasing Core Web Vitals scores by 35%.
- Strengthened **CI/CD pipelines** using GitHub Actions, Jenkins, and Docker, reducing deployment failures by 30% and cutting release time by 40%.

SOFTWARE ENGINEER
Charles Schwab - Phoenix, AZ

Apr 2022 - May 2023

- Designed and implemented an internal API registry using **React** and **TypeScript**, with **production-grade service interface design** adopted by 15+ engineering teams, reducing API discovery time by 50%.
- Engineered **Node.js / Express** backend APIs integrated with **MSSQL** and Redis, applying **data modeling** best practices and reducing backend response times by 35%.
- Implemented **GraphQL** queries and mutations using **schema-driven design**, reducing over-fetching by 40% and improving **production** frontend data efficiency.
- Implemented **OAuth2 / OIDC** authentication, eliminating legacy session vulnerabilities and achieving 100% security compliance.
- Integrated Dockerized workflows into **CI/CD pipelines** (Jenkins, GitHub Actions), reducing manual deployment effort by 60%.

SOFTWARE ENGINEER
American Express - Phoenix, AZ

Apr 2020 - Apr 2022

- Developed AI marketplace web application enabling 10K+ internal users to explore and publish AI models using **React**, **Node.js**, **Python**, and **PostgreSQL** — leveraging **Python** for backend data processing and API integrations.
- Engineered an internal **NLP-to-SQL** application using **Python** and **Node.js**, enabling business users to generate and execute queries, reducing manual SQL effort by 70% and improving analytics turnaround by 45%.

- Designed **production-grade** full-stack applications with **relational database schema design** using **PostgreSQL** and **Okta** authentication, maintaining 99.9% uptime.
- Implemented SEO and performance optimizations using Next.js, improving search visibility by 25%.

APPLICATION PROGRAMMER
American Express - Phoenix, AZ

Jan 2018 - Apr 2020

- Developed Chatbot widget for web application enabling package publication to internal registry, reducing su=
- Designed and implemented single-page applications using **React**, Redux, and **Node.js**, serving 50K+ monthly internal users on **AWS** (EC2, S3).
- Migrated legacy class-based components to functional components with React Hooks, reducing bundle size by 18%.
- Deployed and operated applications on **AWS** (EC2, S3), collaborating with DevOps teams and improving deployment speed by 40%.

Education

University of Bridgeport - Bridgeport, CT | Master of Science
Computer Science, 08/2017

Websites, Portfolios, Profiles

- www.linkedin.com/in/nikhilch07
- www.github.com/nikhilch07
- www.nikhilcheriyala.online